

Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 4: Germination: Conditions Necessary and Types (Hypogeal vs Epigeal) in Maize H614 and Rosecoco Beans

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	D	C	B	B	C	A

USE AFTER: PREDICT PHASE — Predict: individual prediction of what a healthy, non-dormant seed needs and the four-jar predictions (Think-Pair-Share)

Q1. Mr. Wafula's healthy, non-dormant maize and bean seeds were planted in the same field after the same rain, but some still failed to germinate. Which set lists the conditions a healthy, non-dormant seed needs in order to germinate?

- A. Water, oxygen and a suitable temperature**
- B. Oxygen, soil and fertiliser
- C. Light, water and soil
- D. Water, light and a suitable temperature

Correct answer: A. A viable, non-dormant seed needs water, oxygen and a suitable temperature. Light and soil are common misconceptions: the seeds germinated in the dark and soil is not required.

USE AFTER: OBSERVE PHASE — Part A: examining Fatuma's prepared germination jars and identifying changed and controlled variables

Q2. Fatuma set up four jars, each with 20 Rosecoco bean seeds: water + air + 25 °C; dry; no air (boiled cooled water with an oil layer); and wet but kept at 4 °C. Which of these was kept the SAME in all four jars so that the test was fair?

- A. The oxygen supply
- B. The temperature
- C. The water supply
- D. The number of seeds (20) and the seed variety**

Correct answer: D. Each jar changed only one condition (water, air or temperature). The seed variety (Rosecoco) and the number of seeds (20) stayed the same, so differences in germination can be linked to the changed condition.

USE AFTER: OBSERVE PHASE — Part A: counting germinated seeds (radicle visible) and recording them in a table

Q3. A group is counting germinated Rosecoco bean seeds in a jar on Day 5. When should a seed be counted as GERMINATED?

- A. When the first leaves open
- B. When the seed coat becomes soft and swollen
- C. When the radicle has come out of the seed**
- D. When the shoot appears above the soil

Correct answer: C. Germination is counted when the radicle emerges and the embryo resumes growth. A shoot above the soil is emergence, a different event.

USE AFTER: EXPLAIN PHASE — Part A: CER on conditions (Jar 3 no-air evidence and the probe 'What does that tell us about oxygen?')

Q4. In Fatuma's set-up, the jar with water at 25 °C but no air (boiled cooled water with an oil layer) had only 1 of 20 seeds germinate. In the jar with water, air and 25 °C, 18 of 20 germinated. What does this comparison show?

- A. Water is enough on its own for germination
- B. Oxygen is needed, because the embryo uses it in aerobic respiration to release energy for growth**
- C. Light is needed, because the oil layer blocked light
- D. Oxygen is not needed, because one seed still germinated

Correct answer: B. Water and temperature were suitable in both jars, yet germination fell from 18 to 1 when air was removed. The embryo needs oxygen for aerobic respiration to release energy.

USE AFTER: EXPLAIN PHASE — Part A: linking the no-air jar to the waterlogged low patch on Mr. Wafula's farm

Q5. On Mr. Wafula's farm, 6 bean seeds and 3 maize seeds failed to germinate in a low patch that was waterlogged after 25 mm of rain. What is the best explanation?

- A. The rain washed away the light the seeds needed
- B. Water filled the soil pores, so the seeds got too little oxygen**
- C. The soil became too cold for any seed to absorb water
- D. The seeds became dormant because they had too much water

Correct answer: B. Waterlogged soil has its air spaces filled with water, so the seeds lack oxygen for respiration. This is like the no-air jar.

USE AFTER: EXPLAIN PHASE — Part B: drawing and labelling epigeal germination of Rosecoco and hypogeal germination of maize H614

Q6. Rosecoco beans show epigeal germination, and the cotyledons are carried above the soil. Which part of the seedling makes this happen?

- A. The scutellum grows upwards out of the soil
- B. The coleoptile pierces the soil and lifts the cotyledons
- C. The hypocotyl elongates as an arch and pulls the cotyledons above the soil**
- D. The radicle grows upwards to push the cotyledons out

Correct answer: C. In epigeal germination, the hypocotyl lengthens and forms an arch that lifts the cotyledons above the ground. The coleoptile and scutellum belong to maize (hypogeal), and the radicle grows downwards.

USE AFTER: END OF LESSON — Part B: comparison statement explaining why germination (Day 3 and Day 4 radicle) and emergence (Day 7 and Day 8) are different events

Q7. In the time-lapse results, the bean radicle came out on Day 3 and the bean shoot appeared above the soil on Day 8 (emergence). How many days passed between germination and emergence in the bean?

- A. 5 days**
- B. 3 days
- C. 8 days
- D. 11 days

Correct answer: A. Germination (radicle out) was on Day 3 and emergence on Day 8, so the gap is $8 - 3 = 5$ days. This shows germination and emergence are different events.